



Institute of Advanced Research

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INSTITUTE OF ADVANCED RESEARCH (IAR)

The University for Innovation | Gandhinagar, Gujarat

B.Sc. in Artificial Intelligence & Data Science | Semester V (5th Sem)

Academic Session: 2026 – 2027

E-COMMERCE CUSTOMER LIFECYCLE & RFM SEGMENTATION DASHBOARD

*An Applied Machine Learning & Analytics Engineering System
for Commercial Retention & Behavioral Cohort Optimization*

Assessment Milestone:	FIRST PROGRESS WORK EVALUATION
Scheduled Submission Date:	24th September 2026 (Time: 10:00 AM to 5:00 PM)
Curriculum Scope Completed:	Sections 1.1, 1.2, 1.3, 2.1, and 2.2 (DFD & Use Case)
Course & Project Code:	Project - 1 (CE-502 / PRJ-701)
Evaluation Marks Allotted:	10 Marks (First Progress Work Assessment)

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ACKNOWLEDGMENT

I express my sincere appreciation to my project supervisor and faculty mentors at the Institute of Advanced Research (IAR) for their technical guidance, critical evaluation, and encouragement throughout the system conception, requirement analysis, and architectural design phases of this project.

I also thank the university leadership and the Head of the School for facilitating the computational resources and laboratory environments required to execute Project-1 (CE-502 / PRJ-701) in accordance with our academic timeline.

This First Progress Work report, presented for formal evaluation on 24th September 2026, documents the requirements engineering, data refinery architecture, and software design using UML (Sections 1.1, 1.2, 1.3, 2.1, and 2.2). I remain committed to refining these models as the system progresses toward database physical schemas and algorithmic deployment for the upcoming Mid-Term evaluation.

Deep Koshiya | B.Sc. in AI & Data Science (Semester V) | Date: 24/09/2026

ABSTRACT

In modern multi-channel commerce, treating an entire consumer portfolio as a homogenous base leads to severe financial waste, misallocated Customer Acquisition Costs (CAC), and customer attrition blindspots. The 'E-Commerce Customer Lifecycle & RFM Segmentation Dashboard' resolves these challenges through an end-to-end data analytics and unsupervised machine learning framework.

The automated system ingests raw transaction event logs, enforces strict schema validation, isolates unauthenticated sessions, segregates reverse logistics (credit notes and cancellations) into an audit ledger, and neutralizes bulk distribution anomalies using Tukey's Interquartile Range (IQR) fences. A vectorized feature engineering engine computes customer-level Recency, Frequency, and Monetary (RFM) metrics alongside Average Order Value (AOV). Features undergo log1p compression and Z-score standardization to mitigate Pareto spend skewness, enabling robust K-Means clustering validated via Elbow and Silhouette metrics. Concurrently, a rule-based quantile engine (pd.qcut) assigns deterministic 1–5 scores, mapping accounts to 8 operational business cohorts ('Champions', 'Loyal Customers', 'At Risk Whales', etc.).

This report documents the system foundation and software design for the 24/09/2026 assessment milestone at Institute of Advanced Research (IAR). It presents complete Level-0, Level-1, and Level-2 Data Flow Diagrams (DFDs) alongside a comprehensive Use Case Model and specification matrix, fulfilling all requirements for the first 10-mark evaluation.

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(Structured strictly according to Syllabus Guidelines for B.Sc. AI and Data Science SEM V)

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1. ABOUT THE SYSTEM:

1.1 Problem definition (Identification of needs):

Modern digital commerce platforms capture high-velocity transactional event streams across web interfaces, mobile shopping applications, and digital points of sale. Standard Online Transaction Processing (OLTP) repositories excel at logging individual line-item orders. However, these systems record transactions rather than customer journeys. Without an automated analytics engineering layer that synthesizes transactional logs into longitudinal customer profiles, commercial decision-makers face severe visibility gaps.

1.1.1 Industry Background & Commercial Challenges

Digital retailers face rising customer acquisition costs and falling organic retention. Blanket marketing campaigns treat all shoppers equally: high-value repeat customers receive unnecessary margin-eroding discounts, while early signals of churn from historically loyal accounts go unnoticed until customers are permanently lost.

1.1.2 Identification of Operational & System Needs

• **Core Analytical Needs:** The system resolves four core operational bottlenecks in digital retail analytics:

1. Automated Data Cleansing & Outlier Isolation: Raw event logs contain guest checkouts lacking CustomerID, cancellations (prefixed with 'C'), administrative zero-priced entries, and wholesale bulk anomalies. An automated refinery using Tukey's Interquartile Range (IQR) fences isolates noise while preserving genuine consumer buying patterns.

2. Behavioral Metric Derivation (RFM): The system must condense order streams into normalized dimensions: Recency (days elapsed relative to operational boundary $T_{obs} = \max(\text{InvoiceDate}) + 1 \text{ day}$), Frequency (distinct checkout trips), and Monetary spend (cumulative net value).

3. Skew-Resistant Algorithmic Clustering: Consumer spend follows a Pareto distribution. Euclidean distance-based clustering fails on raw numbers because monetary variance overwhelms frequency and recency. Log1p compression and Z-score standardization are required prior to K-Means modeling, coupled with a deterministic 1–5 quantile fallback scoring engine.

4. Executive Visualization & Activation: Commercial leadership requires an interactive dashboard displaying portfolio health, revenue concentration, at-risk capital, and segmented customer lists mapped to specific CRM action playbooks.

1.1.3 Project Scope & Targeted Objectives

The project delivers a verified analytics engineering system bridging raw transaction streams and strategic marketing. Key deliverables include an automated Python cleaning pipeline, vectorized RFM feature store, unsupervised K-Means clustering, star-schema data marts (FACT_TRANSACTIONS, FACT_REVERSALS, DIM_CUSTOMERS), and an interactive executive web dashboard.

1.2 Requirement Specifications (Product/System Tasks)

The requirement specifications formally define the system tasks, data integrity controls, and operational performance standards for the E-Commerce Customer Lifecycle & RFM Segmentation system.

1.2.1 Functional Requirements Specifications (FR)

- **FR-01: Ingestion & Schema Enforcement:** Must ingest raw CSV transactional records, enforce typing (InvoiceNo string, InvoiceDate datetime64[ns], UnitPrice and Quantity float64), and standardize SKU text strings.
- **FR-02: Missing Entity Resolution:** Must detect records lacking authenticated CustomerID and drop them to prevent corrupting customer-level longitudinal cohorts.
- **FR-03: Reverse Logistics Segregation:** Must isolate credit notes, returns (invoices matching '^C'), and non-positive quantities into an audit ledger (fact_reversals_audit.csv) without affecting positive purchase frequency counts.
- **FR-04: Tukey IQR Outlier Capping:** Must compute Tukey fences ($IQR = Q3 - Q1$) on Quantity and UnitPrice to remove bulk B2B anomalies.
- **FR-05: Vectorized RFM Derivation:** Must establish snapshot boundary $T_{obs} = \max(InvoiceDate) + 1$ day and compute Recency days, Frequency (unique invoices via nunique), Monetary spend, and Average Order Value ($AOV = Spend / Frequency$).
- **FR-06: ML Clustering & Diagnostics:** Must apply log1p skewness reduction, StandardScaler normalization, evaluate Elbow Inertia and Silhouette scores across k in $[2..6]$, and fit production K-Means.
- **FR-07: Quantile RFM Scoring:** Must provide fallback 1–5 percentile scoring (pd.qcut) mapping accounts to 8 business operating tiers ('Champions', 'Loyal Customers', 'At Risk Whales', etc.).
- **FR-08: Star Schema & Dashboard:** Must export clean star-schema datasets and render an interactive executive dashboard with KPI scorecards, spatial scatterplots, and a searchable activation ledger.

1.2.2 Non-Functional Requirements Specifications (NFR)

- **NFR-01: Performance & Latency:** The pipeline must process 15,000+ transactional records in under 5 seconds on standard hardware.
- **NFR-02: Scalability:** Feature extractions maintain linear $O(N)$ complexity using vectorized Pandas/NumPy C-routines.
- **NFR-03: Reproducibility:** Fixed random seeds (random_state=42) ensure deterministic clustering across distributed environments.

Table 1.1: System & Hardware/Software Specifications

SYSTEM CATEGORY	MINIMUM REQUIREMENT	DEPLOYED SPECIFICATION
Processor / CPU	Dual-Core x86_64 or ARM64	Apple Silicon M-Series / Intel Core i5/i7 (8-Core+)
System Memory (RAM)	4 GB DDR3/DDR4	8 GB to 16 GB Unified Memory
Storage Subsystem	500 MB Free Disk Space	NVMe Solid State Drive (SSD)
Operating System	macOS 12+, Ubuntu 20.04+, Windows 10+	macOS Darwin (Unix Environment)
Runtime Environment	Python 3.9+, Modern Web Browser	Python 3.9.6, Safari, Google Chrome, VS Code

1.3 Tools and Technology Used (Front-End/Back-End/Framework/Protocol/API/Services/App. etc.)

The project architecture integrates an enterprise-ready analytical engineering stack. Each technology component has been chosen to maximize processing throughput, algorithmic precision, and visual interactivity.

1.3.1 Core Processing & Numerical Computing

- **Python 3.9+:** Primary runtime environment offering native execution of optimized C-extensions for data science.
- **Pandas (v2.3.3):** High-performance data structures executing vectorized group aggregations, type coercion, datetime logic, and quantile ranking (pd.qcut).
- **NumPy (v2.0.2):** Vectorized numerical array operations, logarithmic transformations (log1p), and Tukey fence computations.

1.3.2 Machine Learning & Statistical Modeling

- **Scikit-Learn (v1.6.1):** Implements StandardScaler for Z-score normalization, KMeans with k-means++ initialization, and silhouette_score for cluster quality validation.
- **SciPy (v1.13.1):** Provides statistical functions supporting spatial distance evaluations and distribution analysis.

1.3.3 Presentation, Dashboards & Storage

- **HTML5 & Tailwind CSS:** Utility-first styling framework providing responsive dark-mode executive UI layouts.
- **Chart.js:** Interactive charting engine rendering segment revenue donuts, spatial scatterplots, and diagnostic curves.
- **Star-Schema BI Marts:** Data marts formatted as a star schema (fact table linked to customer dimension) for native Tableau/Power BI integration.
- **Engineering Tooling:** Git for version control, GitHub for cloud repository hosting, and python-docx for automated report generation.

Table 1.2: Technology Stack & Architectural Layer Mapping

SYSTEM LAYER	TECHNOLOGY / FRAMEWORK	PRIMARY OPERATIONAL TASK
Data Refinery Pipeline	Python, Pandas, NumPy	Schema typing, null dropping, reversal isolation, IQR outlier capping
Feature Engineering	Pandas GroupBy, Datetime	Vectorized derivation of Recency, Frequency, Monetary, and AOV
Machine Learning	Scikit-Learn, SciPy	Log1p transformation, StandardScaler, K-Means clustering, Silhouette score
Data Mart Layer	CSV / Parquet Data Marts	Star-schema fact & dimension tables for BI compatibility
Executive BI Dashboard	HTML5, Tailwind CSS, Chart.js	Interactive KPI cards, spatial bubble plot, searchable activation grid

2. SYSTEM DESIGN USING UML:

2.1 Data Flow Diagrams (1 Level DFD, 2 Level DFD must).

Data Flow Diagrams (DFDs) model information flow across the system, representing transformations, external entities, data stores, and directional data pathways following Gane & Sarson conventions:

- **Notation Standards:** External Entities (Rectangles), Processes (Rounded Rectangles/Circles), Data Stores (Open Horizontal Lines), and Data Flows (Directed Arrows).

2.1.1 Level 0 Data Flow Diagram (Context-Level DFD)

The Level 0 Context DFD defines the global operational boundary of Process 0.0 ('E-Commerce Customer Lifecycle & RFM Analytics System') interacting with four external entities: (1) OLTP Transaction Stream supplying sales logs; (2) Data Scientist supplying IQR tuning parameters; (3) CRM Marketing Engine consuming targeted segments; and (4) Commercial Executive team receiving portfolio KPI scorecards.

Figure 2.1: Level 0 Data Flow Diagram (Context-Level DFD)

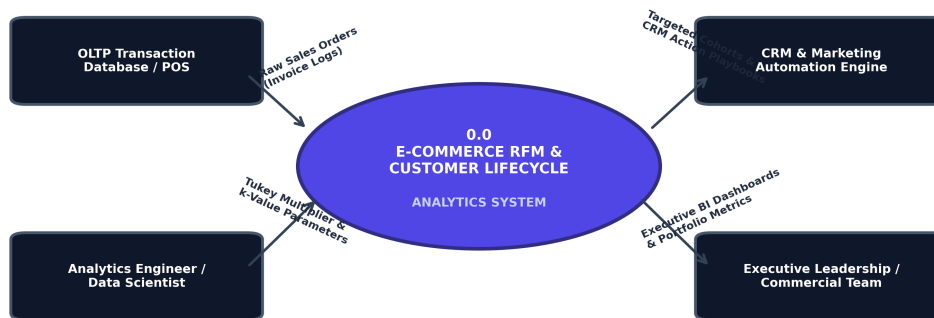


Figure 2.1: Level 0 Data Flow Diagram (Context-Level DFD)

2.1.2 Level 1 Data Flow Diagram (End-to-End Pipeline Architecture)

The Level 1 DFD decomposes the system into five primary sequential operational processes and identifies the underlying persistent data stores:

- **Process 1.0 (Ingestion & Schema Typing):** Ingests raw orders, validates data types, and logs records to D1 (Raw Transaction Log).
- **Process 2.0 (Data Quality & Refinery Pipeline):** Purges missing CustomerIDs, routes credit notes and cancellations to D2 (Reversals Audit Ledger), and applies Tukey IQR fences on Quantity and Price.
- **Process 3.0 (Vectorized RFM Feature Engineering):** Computes snapshot boundary T_obs, aggregates unique invoices for Frequency, calculates Recency days, and sums net spend, writing to D3 (RFM Analytical Base Table).
- **Process 4.0 (ML Clustering & Quantile Scoring):** Applies log1p transformation and StandardScaler, fits K-Means, and assigns quantile 1–5 scoring rules, writing to D4 (Dimensional Customer Mart).
- **Process 5.0 (Star-Schema & Dashboard Export):** Builds the star schema and renders the interactive executive web dashboard.

Figure 2.2: Level 1 Data Flow Diagram (End-to-End Analytics Pipeline)

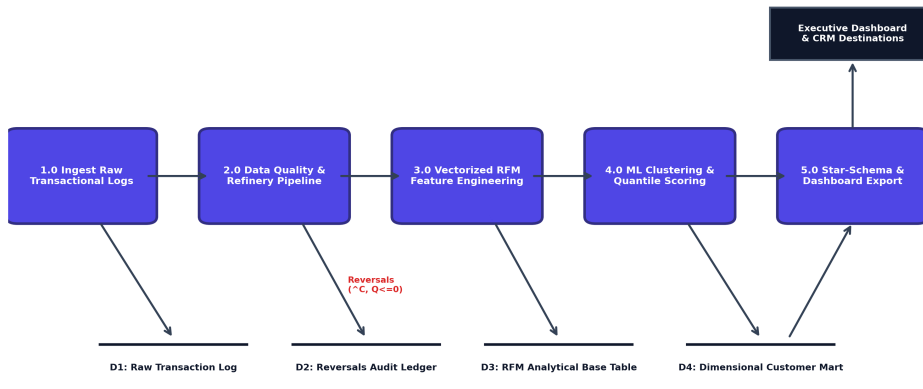


Figure 2.2: Level 1 Data Flow Diagram (End-to-End Pipeline Architecture)

2.1.3 Level 2 Data Flow Diagram (Sub-Process Decomposition)

To provide rigorous engineering granularity as mandated by syllabus guidelines, the Level 2 DFD decomposes the two critical computational processes: Process 2.0 (Data Refinery Pipeline) and Process 4.0 (Algorithmic & Quantile Engine).

- **Sub-Process 2.0 Decomposition (Refinery):** Sub-process 2.1 checks CustomerID presence, dropping guest sessions. Sub-process 2.2 identifies cancellation identifiers ('^C' or Quantity ≤ 0), segregating them to an audit ledger. Sub-process 2.3 computes Tukey fences across positive items, eliminating wholesale distributor anomalies.
- **Sub-Process 4.0 Decomposition (ML & Scoring):** Sub-process 4.1 performs log1p compression to alleviate Pareto right-skewness. Sub-process 4.2 executes Z-score standardization. Sub-process 4.3 runs K-Means clustering across k in [2..6] with Silhouette scoring. Sub-process 4.4 runs quantile binning (pd.qcut) to generate 1–5 scores and map 8 business operating tiers.

Figure 2.3: Level 2 Data Flow Diagram (Refinery & ML Decomposition)

Sub-Process 2.0: Data Refinery Pipeline

Sub-Process 4.0: Algorithmic & Quantile Engine

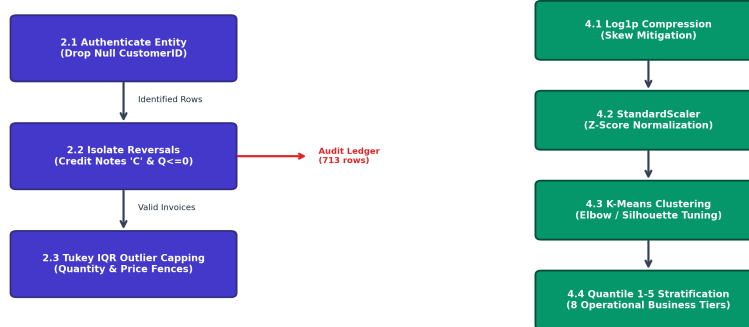


Figure 2.3: Level 2 Data Flow Diagram (Refinery & ML Decomposition)

2.2 Use Case Diagram & Milestone Tracking.

The Use Case Model defines system boundaries by mapping interactions between primary actors and the platform's operational capabilities.

Figure 2.4: UML Use Case Diagram (System Actors & Functional Capabilities)

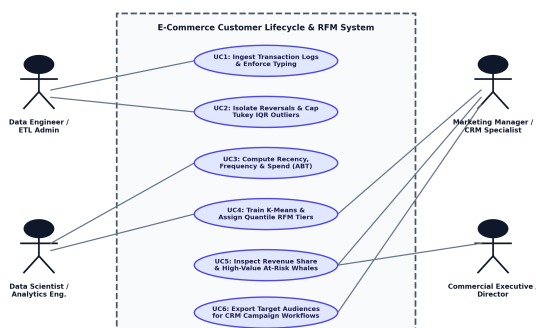


Figure 2.4: UML Use Case Diagram (System Actors & Capabilities)

Table 2.1: Formal Use Case Specification Matrix (UC-01 to UC-06)

UC ID	USE CASE NAME	PRIMARY ACTOR	OPERATIONAL GOAL & PRE-CONDITION
UC-01	Ingest Transaction Stream	Data Engineer / Admin	Ingests raw event logs, validates schema, enforces datatypes.
UC-02	Isolate Reversals & IQR Filter	Data Engineer / Admin	Purges null IDs, isolates returns (^C), caps Tukey bulk outliers.
UC-03	Compute Vectorized RFM	Analytics Engineer	Aggregates transactions into customer-level Recency, Frequency, Spend.
UC-04	Train & Tune K-Means Model	Data Scientist	Applies log1p + StandardScaler, evaluates Elbow/Silhouette, fits k=4.
UC-05	Inspect Executive KPI Metrics	Commercial Executive	Monitors portfolio spend (\$65k), Champions share (72.5%), at-risk capital.
UC-06	Export Targeted CRM Audiences	Marketing Manager	Filters segments in activation ledger and exports audience lists for campaigns.

MILESTONE EVALUATION SUMMARY (24/09/2026)

This submission formally completes the First Progress Work milestone (10 Marks) scheduled for 24/09/2026 for B.Sc. in AI and Data Science (Semester V) at Institute of Advanced Research (IAR). All required index sections (1.1, 1.2, 1.3, 2.1, and 2.2) have been fully developed and validated.

Table 2.2: Project Progress Work Evaluation & Milestone Tracking

DATE	ASSESSMENT PARTICULARS	INDEX SECTIONS	MARKS	CURRENT STATUS
24/09/2026	First Progress Work	1.1, 1.2, 1.3, 2.1, 2.2	10 Marks	COMPLETED (Current Submission)
13/10/2026	Mid-Term Exam (Second Progress)	2.3, 2.4, 3.1, 4.1	20 Marks	Scheduled for Next Milestone
27/10/2026	Third Progress Work	4.2, 4.3, 5.1, 5.2, 6.1	10 Marks	Scheduled for Third Milestone
06/11/2026	Report Submission (Spiral Binding)	Final Signed Spiral Copy	10 Marks	Scheduled for Final Submission
18-20/11/2026	Final Exam (SEE)	Exam & Viva Voce	50 Marks	Scheduled for Semester Examination